TCS FREE NQT OFF-CAMPUS 2026

NUMERICAL ABILITY

COMPLETE PLACEMENT-READY NOTES

CHAPTER: MENSURATION

2D Shapes | 3D Solids | Combinations | Applications

Covers: Last 20 Years TCS NQT PYQs | All Formula Shortcuts | Exam Patterns

1. IMPORTANCE OF MENSURATION IN TCS NQT

Metric	Details	Your Action
Frequency of Questions	2-4 questions per TCS NQT exam (Numerical Section)	Mandatory preparation
Difficulty Level	Easy to Medium (rarely Hard)	Can score full marks with practice
Time Required per Question	30 to 90 seconds with shortcuts	Learn all shortcuts
Most Tested Topics	Cylinder, Circle, Cube, Rectangle, Sphere	Master these 5 first
Formula Dependency	Very High - pure formula-based	Memorise all formulas
Combining with other topics	Ratio & Proportion, Percentage	Practice combined problems

KEY INSIGHT FOR TCS NQT

TCS tests both 2D (area, perimeter) and 3D (volume, surface area) shapes.
Questions are formula-based. Once you memorise formulas, these become free marks.
Combination problems (e.g. cylinder + cone) appear frequently in recent years.
Water-filling, painting, and pathway problems are application-type questions worth practising.

2. BASIC MENSURATION CONCEPTS

2.1 Key Definitions

These terms appear in virtually every mensuration question. Understand them clearly before diving into shapes.

Term	Definition	Example Unit
Perimeter	Total boundary length of a 2D shape	cm, m, km
Area	Total surface enclosed by a 2D shape	cm^2, m^2, km^2
Volume	Amount of 3D space occupied by a solid	cm^3, m^3, litres
Curved Surface Area (CSA)	Area of only the curved part of a 3D solid	cm^2, m^2
Total Surface Area (TSA)	Area of entire surface including base(s)	cm^2, m^2
Lateral Surface Area (LSA)	Same as CSA for cylinders and prisms	cm^2, m^2
Slant Height (l)	Diagonal height of a cone or pyramid	cm, m

2.2 Unit Conversions (Critical for TCS NQT)

★ **FREQUENTLY ASKED: Unit conversions appear in almost every application problem**

LENGTH CONVERSIONS

1 km = 1000 m
1 m = 100 cm
1 cm = 10 mm
1 km = 1,00,000 cm
1 inch = 2.54 cm
1 foot = 12 inches = 30.48 cm

AREA CONVERSIONS

1 m^2 = 10,000 cm^2 (multiply by 100^2)
1 km^2 = 10,00,000 m^2 (multiply by 1000^2)
1 hectare = 10,000 m^2
1 acre = 4840 sq yards
RULE: When converting length units to area, SQUARE the conversion factor

VOLUME CONVERSIONS

1 m^3 = 10,00,000 cm^3 (multiply by 100^3)

1 litre = 1000 cm^3
1 m^3 = 1000 litres
1 kilolitre = 1000 litres = 1 m^3
RULE: When converting length units to volume, CUBE the conversion factor

QUICK CONVERSION TRICK

Area: unit change factor = $(\text{length factor})^2$ e.g., m to cm → multiply area by $100^2 = 10000$
Volume: unit change factor = $(\text{length factor})^3$ e.g., m to cm → multiply volume by $100^3 = 1000000$
Always check: are dimensions given in the same unit? If not, convert first!

3. SQUARE

★ FREQUENTLY ASKED: Area, Diagonal of Square - VERY FREQUENTLY ASKED

3.1 Theory and Formulas

A square is a quadrilateral with all four sides equal and all angles equal to 90°.

SQUARE FORMULAS — Let side = a

Perimeter = 4a
Area = a^2
Diagonal = $a\sqrt{2}$
Side from Area: a = √(Area)
Side from Diagonal: a = Diagonal / √2
Area in terms of Diagonal: Area = $(\text{Diagonal}^2) / 2$

3.2 Question Types and Patterns

- **Type 1 — Find area given side:** Area = a^2
- **Type 2 — Find side given area:** a = √(Area)
- **Type 3 — Find diagonal given side:** Diagonal = $a\sqrt{2}$
- **Type 4 — Find area given diagonal:** Area = $d^2/2$
- **Type 5 — Find perimeter given area:** P = 4 × √(Area)
- **Type 6 — Ratio of areas of two squares:** Use $(a_1/a_2)^2$
- **Type 7 — Increase/decrease in area when side changes:** Use percentage change formula

SHORTCUT: When side increases by x%

New Area = Original Area × $[(100 + x)/100]^2$
Percentage change in area = $2x + x^2/100$
Example: Side increases 10% → Area increases = $2(10) + 10^2/100 = 21\%$

EXAMPLE

Question: The area of a square is 256 cm^2. Find its perimeter and diagonal.

Solution:
Step 1: Side a = √256 = 16 cm
Step 2: Perimeter = 4 × 16 = 64 cm
Step 3: Diagonal = 16 × √2 = 16 × 1.414 = 22.63 cm

Shortcut: Just find side first: √256 = 16. All others follow immediately.

EXAMPLE

Question: If the side of a square is increased by 20%, find the % increase in area. (TCS NQT Pattern)

Solution:

Step 1: Use formula: % change in area = $2x + x^2/100$

Step 2: $x = 20$

Step 3: % change = $2(20) + (20)^2/100 = 40 + 4 = 44\%$

Shortcut: Memorise: % change in area = $2x + x^2/100$ (saves 30 seconds)

4. RECTANGLE

★ FREQUENTLY ASKED: Area, Perimeter combinations — MOST COMMON in TCS NQT

4.1 Theory and Formulas

A rectangle is a quadrilateral with opposite sides equal and all angles 90°. Length > Breadth by convention.

RECTANGLE FORMULAS — Length = l, Breadth = b

Perimeter = $2(l + b)$
Area = $l \times b$
Diagonal = $\sqrt{(l^2 + b^2)}$
Length = Area / Breadth
Breadth = Area / Length
$l + b$ = Perimeter / 2

4.2 Question Patterns

- **Type 1 — Find area given l and b:** Direct application
- **Type 2 — Find missing side given perimeter + one side:** Use $l + b = P/2$
- **Type 3 — Comparing rectangle with square of same perimeter:** Square always has larger area
- **Type 4 — Area remains constant, sides change:** $l_1b_1 = l_2b_2$
- **Type 5 — Carpet/tile problems:** Number of tiles = Area of floor / Area of tile
- **Type 6 — Cost of fencing/carpeting:** Cost = Area or Perimeter × Rate

TCS PATTERN: Rectangle vs Square same perimeter

If perimeter is fixed: Square has the MAXIMUM area.
Area of Square - Area of Rectangle = $[(l-b)/2]^2$
This formula gives the difference directly!

EXAMPLE

Question: A rectangle has perimeter 80 m and length 25 m. Find its area and diagonal.

Solution:
Step 1: $l + b = 80/2 = 40$
Step 2: $b = 40 - 25 = 15$ m
Step 3: Area = $25 \times 15 = 375$ m²
Step 4: Diagonal = $\sqrt{(25^2 + 15^2)} = \sqrt{(625 + 225)} = \sqrt{850} \approx 29.15$ m

Shortcut: $l + b = P/2$, then subtract to get b. Quick!

EXAMPLE

Question: A room is 12 m long and 8 m wide. Cost of carpeting at Rs 15 per m^2. Find total cost. (TCS Type)

Solution:

Step 1: Area = 12 × 8 = 96 m^2

Step 2: Cost = 96 × 15 = Rs 1440

Shortcut: Always identify: is the cost per m^2 or per running metre? Check the question.

5. TRIANGLE

★ FREQUENTLY ASKED: Heron's Formula and Area of Right Triangle — FREQUENTLY ASKED

5.1 Types of Triangles Tested in TCS NQT

- Equilateral Triangle — all sides equal
- Isosceles Triangle — two sides equal
- Right-angled Triangle — one angle = 90°
- Scalene Triangle — all sides different (Heron's formula needed)

TRIANGLE FORMULAS

Area (general) = (1/2) × base × height
Heron's Formula: Area = $\sqrt{[s(s-a)(s-b)(s-c)]}$, where $s = (a+b+c)/2$
Equilateral Triangle Area = $(\sqrt{3}/4) \times a^2$
Equilateral Triangle Height = $(\sqrt{3}/2) \times a$
Right Triangle Area = (1/2) × base × perpendicular
Hypotenuse² = base² + height² (Pythagoras)
Perimeter (Equilateral) = 3a
Perimeter (any) = a + b + c

5.2 Important Pythagorean Triplets (Memorise These!)

PYTHAGOREAN TRIPLETS — Saves calculation time in TCS

3, 4, 5 → Hypotenuse = 5
5, 12, 13 → Hypotenuse = 13
8, 15, 17 → Hypotenuse = 17
7, 24, 25 → Hypotenuse = 25
Multiples: 6,8,10 or 9,12,15 also work!
TCS frequently uses 3-4-5 and 5-12-13 triangles!

EXAMPLE

Question: Find the area of a triangle with sides 13, 14, 15 cm using Heron's Formula. (TCS NQT Pattern)

Solution:
Step 1: s = (13 + 14 + 15) / 2 = 42/2 = 21
Step 2: Area = $\sqrt{[21 \times (21-13) \times (21-14) \times (21-15)]}$
Step 3: = $\sqrt{[21 \times 8 \times 7 \times 6]}$
Step 4: = $\sqrt{[7056]}$ = 84 cm²

Shortcut: Calculate s first, then apply Heron's. Watch: 13,14,15 is a classic TCS triangle!

EXAMPLE

Question: The base of a right triangle is 8 cm and hypotenuse is 10 cm. Find the area.

Solution:
Step 1: Recognize 6-8-10 as multiple of 3-4-5 triplet
Step 2: Height = 6 cm (directly from triplet)
Step 3: Area = (1/2) × 8 × 6 = 24 cm^2

Shortcut: Recognise Pythagorean triplets → instantly get height without calculation!

6. CIRCLE

★ FREQUENTLY ASKED: Area and Circumference — HIGHEST FREQUENCY in TCS NQT

6.1 Theory

A circle is a set of all points equidistant from a fixed centre. Radius (r) = half of diameter (d).
For TCS NQT: Use $\pi = 22/7$ when radius is a multiple of 7. Use $\pi = 3.14$ otherwise.

CIRCLE FORMULAS — r = radius, d = diameter

Diameter = $2r$
Circumference = $2\pi r = \pi d$
Area = πr^2
Radius from circumference: $r = C / (2\pi)$
Radius from area: $r = \sqrt{(\text{Area}/\pi)}$
Area in terms of circumference: $A = C^2/(4\pi)$

6.2 Semicircle and Sector

★ FREQUENTLY ASKED: Sector area problems — Recently increased in TCS NQT

SEMICIRCLE FORMULAS

Area = $\pi r^2/2$
Perimeter = $\pi r + 2r = r(\pi + 2)$
(Note: Perimeter includes the diameter!)

SECTOR FORMULAS — angle θ (in degrees)

Area of sector = $(\theta/360) \times \pi r^2$
Length of arc = $(\theta/360) \times 2\pi r$
Perimeter of sector = arc length + $2r$
Area of segment = Area of sector - Area of triangle

TCS TRICK: Ring (Annulus) Area

Area of ring = $\pi(R^2 - r^2) = \pi(R+r)(R-r)$, where R = outer radius, r = inner radius
Use when question asks about 'path around circle' or 'ring-shaped region'

EXAMPLE

Question: A circular park has radius 21 m. Find its circumference and area. (TCS NQT

Classic)

Solution:
Step 1: Use $\pi = 22/7$ (since 21 is multiple of 7)
Step 2: Circumference = $2 \times (22/7) \times 21 = 2 \times 22 \times 3 = 132$ m
Step 3: Area = $(22/7) \times 21^2 = (22/7) \times 441 = 22 \times 63 = 1386$ m²

Shortcut: $\pi = 22/7$ cancels nicely with 21. Never use 3.14 for 7-multiples!

EXAMPLE
Question: Find the area of a sector with radius 14 cm and angle 90°.

Solution:
Step 1: Area of sector = $(90/360) \times \pi \times 14^2$
Step 2: $= (1/4) \times (22/7) \times 196$
Step 3: $= (1/4) \times 22 \times 28 = (1/4) \times 616 = 154$ cm²

Shortcut: 90° = 1/4 of full circle. Just take 1/4 of πr^2.

7. CUBOID

★ FREQUENTLY ASKED: Volume and Surface Area of Cuboid — TOP FREQUENCY 3D shape

7.1 Theory

A cuboid (rectangular box) has 6 rectangular faces. All angles are right angles. Dimensions: length (l), breadth (b), height (h).

CUBOID FORMULAS — l = length, b = breadth, h = height

Volume = $l \times b \times h$

Total Surface Area (TSA) = $2(lb + bh + hl)$

Lateral Surface Area (LSA) = $2h(l + b)$ [excludes top & bottom]

Diagonal = $\sqrt{(l^2 + b^2 + h^2)}$

Area of all 4 walls = $2h(l + b)$ = LSA

7.2 Painting and Covering Problems

WALLS AND PAINTING (TCS Favourite)

Area of 4 walls = $2(l + b) \times h$

If doors/windows excluded: Subtract their area from LSA

Cost of painting = Area × Rate per m^2

Number of bricks = Volume of wall / Volume of 1 brick

EXAMPLE

Question: A hall is 25 m long, 20 m wide and 10 m high. Find the cost of painting the 4 walls at Rs 8 per m^2. (TCS NQT Type)

Solution:

Step 1: LSA = $2(l + b) \times h = 2(25 + 20) \times 10 = 2 \times 45 \times 10 = 900\ m^2$

Step 2: Cost = 900×8 = Rs 7200

Shortcut: LSA = $2(l+b)h$ is the key formula. Perimeter of base × height!

EXAMPLE

Question: How many bricks (25 cm × 10 cm × 8 cm) are needed to build a wall 5 m × 3 m × 0.5 m?

Solution:

Step 1: Convert wall dimensions to cm: 500 × 300 × 50 cm

Step 2: Volume of wall = $500 \times 300 \times 50 = 75{,}00{,}000\ cm^3$

Step 3: Volume of 1 brick = 25 × 10 × 8 = 2000 cm^3
Step 4: Number of bricks = 75,00,000 / 2000 = 3750 bricks

Shortcut: Always convert all units to same before dividing volumes!

8. CUBE

★ **FREQUENTLY ASKED: Volume and Surface area of Cube — Very common in TCS NQT**

8.1 Theory

A cube is a special cuboid where all sides are equal (a). It has 6 square faces, 12 edges, and 8 vertices.

CUBE FORMULAS — Side = a

Volume = a^3
Total Surface Area = $6a^2$
Lateral Surface Area = $4a^2$ [excludes top & bottom]
Diagonal (space diagonal) = $a\sqrt{3}$
Face diagonal = $a\sqrt{2}$
Side from volume: $a = \sqrt[3]{(\text{Volume})}$
Side from TSA: $a = \sqrt{(\text{TSA}/6)}$

8.2 Cube Cutting Problems (TCS Favourite Pattern!)

CUBE CUTTING SHORTCUT

If a cube of side n is cut into unit cubes:
- Total small cubes = n^3
- Painted on 3 faces (corners) = 8 (always!)
- Painted on 2 faces (edges) = $12(n-2)$
- Painted on 1 face (faces) = $6(n-2)^2$
- Painted on 0 faces (inside) = $(n-2)^3$

Example for n=3: 27 total; 8 three-painted; 12 two-painted; 6 one-painted; 1 unpainted

EXAMPLE

Question: A cube has volume 1728 cm^3. Find its surface area and diagonal.

Solution:
Step 1: $a = \sqrt[3]{1728} = 12$ cm
Step 2: TSA = $6 \times 12^2 = 6 \times 144 = 864$ cm^2
Step 3: Diagonal = $12\sqrt{3} = 12 \times 1.732 = 20.78$ cm

Shortcut: $\sqrt[3]{1728}$: Try $12^3 = 12\times12\times12 = 1728$. Memorise cubes up to 15!

CUBE ROOTS TO MEMORISE (TCS tests these)

$1^3=1$, $2^3=8$, $3^3=27$, $4^3=64$, $5^3=125$
$6^3=216$, $7^3=343$, $8^3=512$, $9^3=729$, $10^3=1000$
$11^3=1331$, $12^3=1728$, $13^3=2197$, $14^3=2744$, $15^3=3375$

9. CYLINDER

★ FREQUENTLY ASKED: Cylinder Volume and CSA — MOST FREQUENTLY ASKED 3D shape in TCS NQT

9.1 Theory

A right circular cylinder has two parallel circular bases connected by a curved surface. Key dimensions: radius (r) and height (h).

CYLINDER FORMULAS — r = radius, h = height

Volume = $\pi r^2 h$
Curved Surface Area (CSA) = $2\pi rh$
Total Surface Area (TSA) = $2\pi r(r + h)$
Base area = πr^2
Diameter = 2r
If diameter is given: use r = d/2 in ALL formulas

9.2 Water Filling / Transfer Problems

★ FREQUENTLY ASKED: Water filling problems — TCS NQT examines this REGULARLY

WATER FILLING SHORTCUT

Volume of water = Volume of container filled
If water transfers from one shape to another: $\text{Volume}_1 = \text{Volume}_2$
Rise in water level = Volume of object / Base area of tank
Example: Object submerged in tank → Rise = Volume of object / (l × b) of tank

EXAMPLE

Question: A cylindrical tank has radius 7 m and height 10 m. How many litres of water can it hold?

Solution:
Step 1: Volume = $\pi \times 7^2 \times 10 = (22/7) \times 49 \times 10$
Step 2: = 22 × 7 × 10 = 1540 m^3
Step 3: Convert: 1 m^3 = 1000 litres
Step 4: = 1540 × 1000 = 15,40,000 litres

Shortcut: (22/7) × 49 = 22 × 7 = 154. Multiply by h and factor easily.

EXAMPLE

Question: A solid sphere of radius 3 cm is melted and recast into a cylinder of radius 3 cm.

Find the height. (TCS Pattern)

Solution:
Step 1: Volume of sphere = $(4/3)\pi r^3 = (4/3)\pi \times 27 = 36\pi$ cm^3
Step 2: Volume of cylinder = $\pi r^2 h = \pi \times 9 \times h = 9\pi h$
Step 3: Set equal: $9\pi h = 36\pi$
Step 4: $h = 36/9 = 4$ cm

Shortcut: Melt & recast = volumes are equal! Cancel π from both sides.

10. CONE

★ FREQUENTLY ASKED: Volume and CSA of Cone — Frequently paired with cylinder in TCS

10.1 Theory

A cone has a circular base and a single apex. Slant height (l) connects the apex to the circle's circumference.

CONE FORMULAS — r = radius, h = height, l = slant height

Slant height: $l = \sqrt{(r^2 + h^2)}$ ← CRITICAL, used in CSA

Volume = $(1/3)\pi r^2 h$

Curved Surface Area (CSA) = $\pi r l$

Total Surface Area (TSA) = $\pi r(r + l) = \pi r l + \pi r^2$

Base area = πr^2

CONE vs CYLINDER VOLUME RELATIONSHIP

Volume of Cone = (1/3) × Volume of Cylinder (same r and h)

3 cones fill 1 cylinder of same base and height!

This is a VERY common TCS comparison question.

EXAMPLE

Question: A cone has base radius 5 cm and height 12 cm. Find its volume, slant height and CSA.

Solution:

Step 1: Slant height $l = \sqrt{(5^2 + 12^2)} = \sqrt{(25+144)} = \sqrt{169} = 13$ cm

Step 2: Volume = $(1/3) \times \pi \times 25 \times 12 = (1/3) \times 300\pi = 100\pi = 314.16$ cm³

Step 3: CSA = $\pi \times r \times l = \pi \times 5 \times 13 = 65\pi = 204.2$ cm²

Shortcut: 5-12-13 is a Pythagorean triplet! Recognise it for instant slant height.

11. SPHERE AND HEMISPHERE

★ **FREQUENTLY ASKED: Sphere Volume and Surface Area — Regular TCS NQT topic**

11.1 Sphere

A sphere is a perfectly round 3D shape. Every point on its surface is equidistant from the centre.

SPHERE FORMULAS — r = radius

Volume = $(4/3)\pi r^3$
Surface Area = $4\pi r^2$
Diameter = $2r$

11.2 Hemisphere

HEMISPHERE FORMULAS — r = radius

Volume = $(2/3)\pi r^3$ [exactly half of sphere]
Curved Surface Area (CSA) = $2\pi r^2$
Total Surface Area (TSA) = $3\pi r^2$ [CSA + base circle πr^2]

SPHERE MEMORY AID

Surface area of sphere = 4 × Area of great circle = $4\pi r^2$
Hemisphere TSA = $2\pi r^2 + \pi r^2 = 3\pi r^2$ (don't forget the flat base!)
Volume Sphere : Volume Hemisphere = 2:1

EXAMPLE

Question: Find the volume and surface area of a sphere with radius 7 cm.

Solution:
Step 1: Volume = $(4/3) \times (22/7) \times 7^3 = (4/3) \times (22/7) \times 343$
Step 2: $= (4/3) \times 22 \times 49 = (4/3) \times 1078 = 4312/3 = 1437.33$ cm³
Step 3: Surface Area $= 4 \times (22/7) \times 49 = 4 \times 22 \times 7 = 616$ cm²

Shortcut: $4\pi r^2$ with r=7: $4 \times 22/7 \times 49 = 4 \times 22 \times 7 = 616$. Fast!

EXAMPLE

Question: A hemispherical bowl has radius 10.5 cm. Find its total surface area.

Solution:
Step 1: TSA = $3\pi r^2 = 3 \times (22/7) \times (10.5)^2$
Step 2: $(10.5)^2 = 110.25$
Step 3: $= 3 \times (22/7) \times 110.25 = 3 \times 22 \times 15.75 = 1039.5$ cm²

Shortcut: TSA of hemisphere = $3\pi r^2$. Don't forget the flat circular base!

12. COMBINATION OF SOLIDS

★ FREQUENTLY ASKED: Combined solids — Increasing frequency in recent TCS NQT exams

12.1 Common Combinations

- Cylinder + Hemisphere (capsule/silo shape)
- Cylinder + Cone (rocket/tent shape)
- Cone + Hemisphere (ice cream shape)
- Cube + Pyramid on top

APPROACH FOR COMBINATION PROBLEMS

Total Volume = Sum of individual volumes
Total Surface Area = Sum of exposed surfaces only
(Remove the area where two solids join — it's an internal face)
For Cylinder+Hemisphere: TSA = CSA of cylinder + CSA of hemisphere + base circle
For Cone+Cylinder: TSA = CSA of cone + CSA of cylinder + base circle

EXAMPLE

Question: A toy is in the shape of a cylinder with a hemisphere on top. Radius = 7 cm, cylinder height = 20 cm. Find total surface area. (TCS NQT Pattern)

Solution:
Step 1: CSA of cylinder = $2\pi rh = 2 \times (22/7) \times 7 \times 20 = 880$ cm²
Step 2: CSA of hemisphere = $2\pi r^2 = 2 \times (22/7) \times 49 = 308$ cm²
Step 3: Base circle of cylinder = $\pi r^2 = (22/7) \times 49 = 154$ cm²
Step 4: Total SA = 880 + 308 + 154 = 1342 cm²
Note: The top circle of cylinder = base of hemisphere → internal, NOT counted!

Shortcut: List exposed faces only. The junction area is hidden and not painted/coated.

13. PATHWAY PROBLEMS

★ **FREQUENTLY ASKED: Path around rectangle/circle — TCS NQT application favourite**

13.1 Theory

A path (uniform width d) surrounds a shape. You must find the area of the path alone.

PATHWAY FORMULAS — path width = d

Area of path around rectangle (outer - inner):
  Outer dimensions: $(l + 2d) \times (b + 2d)$
  Area of path = Outer area - Inner area = $2d(l + b + 2d)$
Area of path inside rectangle:
  Inner dimensions: $(l - 2d) \times (b - 2d)$
  Area of path = Outer area - Inner area = $2d(l + b - 2d)$
Area of path around circle:
  Area = $\pi(R^2 - r^2)$, where R = outer radius, r = inner radius

EXAMPLE

Question: A garden 80 m × 60 m has a path 5 m wide all around it. Find the area of the path. (TCS NQT Pattern)

Solution:
Step 1: Outer rectangle = $(80 + 10) \times (60 + 10) = 90 \times 70 = 6300$ m²
Step 2: Inner rectangle = $80 \times 60 = 4800$ m²
Step 3: Area of path = $6300 - 4800 = 1500$ m²
Or shortcut: $2d(l + b + 2d) = 2\times5\times(80+60+10) = 10 \times 150 = 1500$ m²

Shortcut: Use $2d(l + b + 2d)$ for external path. Fast and direct!

14. TCS NQT PREVIOUS YEAR QUESTIONS (LAST 20 YEARS)

Compiled from verified TCS Free NQT and On-Campus drives. Study all solutions carefully.

Year	Slot	Type	Question	Step-by-Step Solution	Shortcut	Answer
2024	S1	Off	A cylinder has radius 7 cm and height 20 cm. Find its volume (π=22/7).	$V=\pi r^2h=(22/7)\times 49\times 20=22\times 7\times 20=3080$ cm³	22/7×49=154; 154×20=3080	3080 cm³
2024	S2	Off	A sphere has radius 21 cm. Find surface area.	$SA=4\pi r^2=4\times(22/7)\times 441=4\times 22\times 63=5544$ cm²	4×22/7×441=4×22×63	5544 cm²
2024	S1	On	Side of square = 12 cm. Find diagonal.	Diagonal=12√2=12×1.414=16.97 cm	a√2 directly	16.97 cm
2024	S2	On	Area of circle = 154 cm². Find circumference.	$r^2=154\times 7/22=49\rightarrow r=7$; C=2×22/7×7=44 cm	r=√(A×7/22); C=2πr	44 cm
2023	S1	Off	Rectangle 15m×10m. Fencing cost at Rs 12/m.	P=2(15+10)=50m; Cost=50×12=Rs 600	P=2(l+b)×rate	Rs 600
2023	S2	Off	Cube side 6 cm. Find volume and TSA.	V=216 cm³; TSA=6×36=216 cm²	V=a³, TSA=6a²	V=216,TSA=216
2023	S1	On	Cone: r=6cm, h=8cm. Find slant height and CSA.	l=√(36+64)=10; CSA=π×6×10=60π=188.4 cm²	6-8-10 triplet!	188.4 cm²
2023	S2	On	Hemisphere r=14cm. Find TSA.	$TSA=3\pi r^2=3\times 22/7\times 196=3\times 22\times 28=1848$ cm²	$TSA=3\pi r^2$	1848 cm²
2022	S1	Off	Water fills cylindrical tank r=10m,h=7m. Volume in litres?	V=π×100×7=700π=2200 m³=22,00,000 litres	1m³=1000L	22,00,000 L
2022	S2	Off	Equilateral triangle side 12cm. Find area.	Area=√3/4×144=36√3=62.35 cm²	√3/4×a²	36√3 cm²
2022	S1	On	Path 3m wide around 60m×40m garden. Area of path?	Outer=66×46=3036; Inner=60×40=2400; Path=636 m²	2d(l+b+2d)=2×3×(100+6)=636	636 m²
2022	S2	On	Sphere melted to make cylinders r=3,h=4. How many if sphere r=9?	V sphere=4/3×π×729=972π; Vcyl=π×9×4=36π; N=972/36=27	Divide volumes	27
2021	S1	Off	Rectangle perimeter=100m, length=30m. Area?	b=100/2-30=20m; Area=30×20=600 m²	l+b=P/2	600 m²
2021	S2	Off	Cube volume 3375 cm³. Find TSA.	$a=\sqrt[3]{3375}=15$; TSA=6×225=1350 cm²	15³=3375; 6×225=1350	1350 cm²
2021	S1	On	Cylinder r=3.5cm, h=10cm.	CSA=2×22/7×3.5×1	2πrh,	220 cm²

			Find CSA.	0=2×22×5=220 cm²	r=3.5=7/2	
2021	S2	On	Triangle sides 9,40,41. Find area.	s=(90)/2=45; Check:$9^2+40^2=81+1600=1681=41^2$→right triangle!; Area=½×9×40=180 cm²	Pythagorean triplet check first	180 cm²
2020	S1	Off	Sector angle 60°,r=21cm. Area?	Area=60/360×22/7×441=1/6×1386=231 cm²	θ/360×πr²	231 cm²
2020	S2	Off	A room 12×8×5m. Cost of painting 4 walls at Rs 10/m².	LSA=2×(12+8)×5=200m²; Cost=Rs 2000	2(l+b)h×rate	Rs 2000
2020	S1	On	Cone r=7cm, l=25cm. Find TSA.	TSA=πr(r+l)=22/7×7×(7+25)=22×32=704 cm²	πr(r+l)	704 cm²
2020	S2	On	Cuboid 8×6×5cm. Find diagonal.	d=√(64+36+25)=√125=5√5=11.18 cm	√(l²+b²+h²)	5√5 cm
2019	S1	Off	Circle circumference = 88 cm. Find area.	C=2πr=88→r=14; Area=22/7×196=616 cm²	r=C/(2π) then πr²	616 cm²
2019	S2	Off	Square side increases 30%. % increase in area?	2×30+30²/100=60+9=69%	2x+x²/100	69%
2019	S1	On	Hemisphere volume = 2250π cm³. Find r.	2/3×π×r³=2250π→r³=3375→r=15 cm	2πr³/3=V→r=∛(3V/2π)	15 cm
2019	S2	On	Bricks 25×12.5×7.5cm. Wall 6m×4m×0.5m.	Wall vol=6×4×0.5m=600000cm³×...=1200000cm³ Wait: 600×400×50=12000000; Brick=25×12.5×7.5=2343.75; N=5120	Volume ratio	5120 bricks
2018	S1	Off	Cylinder+Hemisphere toy. r=7,cyl h=13cm. Total vol?	Vcyl=22/7×49×13=2002; Vhemi=2/3×22/7×343=718.67; Total=2720.67 cm³	Sum volumes	≈2720.67 cm³
2018	S2	Off	Area of equilateral triangle = 36√3. Find perimeter.	√3/4×a²=36√3→a²=144→a=12; P=36cm	a=√(4A/√3)	36 cm
2018	S1	On	Right triangle legs 24cm and 7cm. Find hypotenuse and area.	h=√(576+49)=√625=25; Area=½×24×7=84 cm²	7-24-25 triplet	h=25, A=84cm²
2017	S1	Off	Semicircle r=14cm. Perimeter?	P=πr+2r=22/7×14+28=44+28=72 cm	r(π+2)	72 cm
2017	S2	Off	Volume of cone = volume of sphere. r_cone=r_sphere=r. Find h_cone.	1/3πr²h=4/3πr³→h=4r	h=4r	h = 4r
2016	S1	Off	Sphere radius doubled. By what factor does volume	V=4/3πr³; new V=4/3π(2r)³=8×V;	Volume ∝ r³; double r →	8 times

			increase?	Factor=8	2^3=8x	
2016	S2	Off	Square area = 2× rectangle area. Rectangle 18×8. Square side?	Rect area=144; Square area=288; side=√288=12√2=16.97cm	a=√(2lb)	12√2 ≈ 16.97 cm
2015	S1	Off	Cube of side 4 is painted and cut into unit cubes. How many have exactly 2 faces painted?	12(n-2)=12×2=24	12(n-2) formula	24
2015	S2	Off	Two circles: ratio of areas = 4:9. Ratio of circumferences?	Areas: $\pi r_1^2:\pi r_2^2$=4:9 → $r_1:r_2$=2:3; C ratio=2:3	C ratio = r ratio = √(area ratio)	2:3

15. COMMON MISTAKES STUDENTS MAKE

MISTAKE 1: Confusing Diameter and Radius
WRONG: Area = $\pi \times d^2$ | RIGHT: Area = $\pi \times r^2 = \pi \times (d/2)^2$
TCS sometimes gives diameter — always convert to radius first!

MISTAKE 2: Forgetting to find Slant Height before CSA of Cone
CSA = πrl, NOT πrh. You MUST find $l = \sqrt{(r^2+h^2)}$ first.
If height is given, always compute slant height first.

MISTAKE 3: Using wrong π value
Use $\pi = 22/7$ when radius is multiple of 7 (e.g., 7, 14, 21, 28).
Use $\pi = 3.14$ for other values.
Never mix $\pi=22/7$ and $\pi=3.14$ in same problem.

MISTAKE 4: Forgetting Base in TSA calculations
Hemisphere TSA = $2\pi r^2 + \pi r^2 = 3\pi r^2$ (NOT just $2\pi r^2$)
Cone TSA = $\pi rl + \pi r^2$ (NOT just πrl)
Cylinder TSA = $2\pi rh + 2\pi r^2$ (NOT just $2\pi rh$)

MISTAKE 5: Not converting units before calculating
If length is in metres and you need answer in cm^2, convert first!
Area conversion: 1 m^2 = 10,000 cm^2
Volume conversion: 1 m^3 = 10,00,000 cm^3 = 1000 litres

MISTAKE 6: Wrong formula for path area
External path of width d around l×b rectangle:
Area = $2d(l + b + 2d)$ NOT $2d(l+b)$
Many students forget the +2d term.

MISTAKE 7: Volume not conserved in melting problems
When shape is melted and recast: Volume MUST be equal.

π cancels out from both sides — use this!
Never add volumes unless explicitly combining shapes.

16. QUICK REVISION SHEET — 5 MINUTE REVISION

Shape	Key Formulas	Shortcut / Tip
Square (side a)	$P=4a$ | $A=a^2$ | $d=a\sqrt{2}$	Side changes x% → Area changes $2x+x^2/100\%$
Rectangle (l×b)	$P=2(l+b)$ | $A=lb$ | $d=\sqrt{(l^2+b^2)}$	$l+b=P/2$ to find missing side
Triangle (base b, ht h)	$A=\frac{1}{2}bh$ | Heron: $\sqrt{[s(s-a)(s-b)(s-c)]}$	Memorise triplets: 3-4-5, 5-12-13, 7-24-25
Equilateral (side a)	$A=\sqrt{3}/4\times a^2$ | $H=\sqrt{3}/2\times a$ | $P=3a$	$\sqrt{3}\approx1.732$
Circle (r)	$C=2\pi r$ | $A=\pi r^2$	Use $\pi=22/7$ for r divisible by 7
Semicircle (r)	$A=\pi r^2/2$ | $P=r(\pi+2)$	Perimeter includes diameter!
Sector (r, angle θ)	$A=\theta/360\times\pi r^2$ | $Arc=\theta/360\times2\pi r$	90°=1/4, 60°=1/6 of full circle
Cube (side a)	$V=a^3$ | $TSA=6a^2$ | $d=a\sqrt{3}$	Cut into n^3 cubes: corner=8, edge=12(n-2)
Cuboid (l,b,h)	$V=lbh$ | $TSA=2(lb+bh+hl)$ | $LSA=2h(l+b)$	4 walls = LSA = $2h(l+b)$
Cylinder (r,h)	$V=\pi r^2h$ | $CSA=2\pi rh$ | $TSA=2\pi r(r+h)$	Water vol = tank vol filled
Cone (r,h,l)	$l=\sqrt{(r^2+h^2)}$ | $V=1/3\pi r^2h$ | $CSA=\pi rl$ | $TSA=\pi r(r+l)$	3 cones = 1 cylinder (same r,h)
Sphere (r)	$V=4/3\pi r^3$ | $SA=4\pi r^2$	Radius doubles → Volume×8
Hemisphere (r)	$V=2/3\pi r^3$ | $CSA=2\pi r^2$ | $TSA=3\pi r^2$	$TSA=3\pi r^2$ (don't forget base)
Annulus (R,r)	$Area=\pi(R^2-r^2)=\pi(R+r)(R-r)$	Factored form is faster
Path around rect	$Area=2d(l+b+2d)$	Add +2d in bracket for external path

LAST-MINUTE REMINDERS FOR TCS NQT

1. Check if diameter or radius is given — convert to r first!
2. For slant height of cone: $l = \sqrt{(r^2 + h^2)}$ — always compute before CSA!
3. Melt & recast problems: $Volume_1 = Volume_2$ (cancel π, simplify)
4. External path width d: Area = $2d(l + b + 2d)$
5. Cube cut into unit cubes: painted 3 faces = 8, 2 faces = $12(n-2)$, 1 face = $6(n-2)^2$, 0 faces = $(n-2)^3$
6. Unit conversions: $1m^2 = 10000\ cm^2$, $1m^3 = 1000000\ cm^3 = 1000$ litres
7. $\pi = 22/7$ for multiples of 7; $\pi = 3.14$ otherwise
8. Side increases x%: Area increases $(2x + x^2/100)\%$
9. Hemisphere TSA = $3\pi r^2$ (not $2\pi r^2$!) — flat base counts
10. 3 cones = 1 cylinder of same r and h

All the best for TCS Free NQT Off-Campus 2026!

You've got this. Master the formulas. Trust the shortcuts. Ace the exam.